

Practice Questions for 2024-10-22-Convexity

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1 Questions

1. a. Prove that the intersection of two convex sets is also a convex set.
 - A. Let C_1 and C_2 be two convex sets. Let $x, y \in C_1 \cap C_2$. Then $x, y \in C_1$ and $x, y \in C_2$. Since C_1 and C_2 are convex, for any $t \in [0, 1]$, $tx + (1-t)y \in C_1$ and $tx + (1-t)y \in C_2$. Therefore, $tx + (1-t)y \in C_1 \cap C_2$, proving that $C_1 \cap C_2$ is convex.
 - B. The intersection of two convex sets is not necessarily convex. A counterexample can be constructed.
 - C. Convexity is only defined for sets in $\mathbb{R}^2$, so the intersection of two convex sets in $\mathbb{R}^3$ is not necessarily convex.
 - D. The proof requires the use of the Caratheodory theorem.
 - E. The statement is only true if the convex sets are bounded.
 - b. Explain why the convex hull of a set S , denoted $\text{Conv}(S)$, is the smallest convex set containing S .
 - A. Because $\text{Conv}(S)$ is defined as the intersection of all convex sets containing S , any smaller convex set would not contain all of S .
 - B. Because $\text{Conv}(S)$ is defined as the union of all convex sets containing S , it is the largest convex set containing S .
 - C. This is a consequence of Caratheodory's theorem.
 - D. This is only true if S is a finite set.
 - E. The statement is only true if S is a convex set.
 - c. Using Caratheodory's Theorem, explain why any point in the convex hull of a set S in $\mathbb{R}^m$ can be expressed as a convex combination of at most $m + 1$ points from S .
 - A. Caratheodory's Theorem directly states this property; no further explanation is needed.
 - B. The proof involves constructing a linear program and showing that its optimal solution uses at most $m + 1$ points from S .
 - C. This is a consequence of the Separation Theorem.
 - D. This is only true if S is a finite set.
 - E. This is only true if S is a convex set.
2. Prove that the intersection of two convex sets is also a convex set.
 - a. Let C_1 and C_2 be two convex sets. Consider $x, y \in C_1 \cap C_2$. Then $x, y \in C_1$ and $x, y \in C_2$. Since C_1 and C_2 are convex, for any $t \in [0, 1]$, $tx + (1-t)y \in C_1$ and $tx + (1-t)y \in C_2$. Therefore, $tx + (1-t)y \in C_1 \cap C_2$, proving that $C_1 \cap C_2$ is convex.

- b. The intersection of two convex sets is not necessarily convex. A counterexample can be constructed.
 - c. Convexity is only defined for single sets, not intersections of sets.
 - d. Let C_1 and C_2 be two convex sets. The intersection $C_1 \cap C_2$ is convex if and only if C_1 and C_2 are disjoint.
 - e. The intersection of two convex sets is only convex if they are both bounded.
3. Explain why the convex hull of a set S , denoted $\text{Conv}(S)$, is the smallest convex set containing S .
 - a. The convex hull is the largest convex set containing S .
 - b. The convex hull is defined as the intersection of all convex sets containing S . Since it's an intersection of convex sets, it's convex, and it contains S by definition. Any smaller convex set containing S would have to be a subset of this intersection, making $\text{Conv}(S)$ the smallest.
 - c. The convex hull is not necessarily the smallest convex set containing S .
 - d. The convex hull is only the smallest convex set containing S if S is finite.
 - e. The convex hull is only defined for convex sets S .
4. Using Caratheodory's Theorem, explain why any point in the convex hull of a set S in R^m can be expressed as a convex combination of at most $m + 1$ points from S .
 - a. Caratheodory's Theorem states that any point in the convex hull can be expressed as a convex combination of at most m points. The $+1$ is a mistake.
 - b. Caratheodory's Theorem doesn't relate to the number of points in a convex combination.
 - c. Caratheodory's Theorem states that any point in the convex hull can be expressed as a convex combination of at most $m + 1$ points. The proof involves constructing a linear program to find such a combination and showing that a basic feasible solution will have at most $m + 1$ non-zero variables.
 - d. Only points in the interior of the convex hull can be expressed as a convex combination of at most $m + 1$ points.
 - e. The number of points required is unbounded.
5. Consider a set $S = \{(1, 0), (0, 1), (1, 1)\}$ in R^2 . Describe the convex hull of S , $\text{Conv}(S)$, and provide a point within $\text{Conv}(S)$ that is a convex combination of all three points in S .
 - a. $\text{Conv}(S)$ is the triangle with vertices $(1, 0)$, $(0, 1)$, and $(1, 1)$. The point $(2/3, 2/3)$ is a convex combination of all three points.
 - b. $\text{Conv}(S)$ is the line segment connecting $(1, 0)$ and $(0, 1)$. The point $(1/2, 1/2)$ is a convex combination of $(1,0)$ and $(0,1)$.
 - c. $\text{Conv}(S)$ is the set S itself. There is no convex combination of all three points.
 - d. $\text{Conv}(S)$ is the entire R^2 plane.
 - e. $\text{Conv}(S)$ is undefined for this set S .
6.
 - a. Prove that the intersection of two convex sets is also a convex set.

- A. Let C_1 and C_2 be two convex sets. Let $x, y \in C_1 \cap C_2$. Then $x, y \in C_1$ and $x, y \in C_2$. Since C_1 is convex, for any $t \in [0, 1]$, $tx + (1-t)y \in C_1$. Similarly, since C_2 is convex, $tx + (1-t)y \in C_2$. Therefore, $tx + (1-t)y \in C_1 \cap C_2$, proving that $C_1 \cap C_2$ is convex.
 - B. The intersection of two convex sets is not necessarily convex. A counterexample can be constructed.
 - C. Convexity is only defined for sets in R^2 , so the intersection of two convex sets in R^3 is not necessarily convex.
 - D. The intersection of two convex sets is only convex if the sets are disjoint.
 - E. The statement is only true for specific types of convex sets, such as polyhedra.
- b. Explain why the convex hull of a set S , denoted $\text{Conv}(S)$, is the smallest convex set containing S .
- A. By definition, $\text{Conv}(S)$ is the intersection of all convex sets containing S . Since it's an intersection of convex sets, it's convex. Any smaller convex set would necessarily exclude some points from S , contradicting the definition.
 - B. The convex hull is the largest convex set containing S .
 - C. The convex hull is only the smallest convex set containing S if S is finite.
 - D. The convex hull is not necessarily a convex set.
 - E. The concept of "smallest" is not well-defined for sets.
- c. Using Caratheodory's Theorem, explain why any point in the convex hull of a set S in R^m can be expressed as a convex combination of at most $m + 1$ points from S .
- A. Caratheodory's Theorem directly states that any point in the convex hull of a set S in R^m can be written as a convex combination of at most $m + 1$ points from S . The proof involves constructing a linear program and using the properties of basic feasible solutions.
 - B. This is only true if the set S is convex.
 - C. Caratheodory's Theorem only applies to finite sets S .
 - D. The number of points needed is at most m , not $m + 1$.
 - E. The statement is not related to Caratheodory's Theorem.

2 Answers

1.
 - a. Prove that the intersection of two convex sets is also a convex set.
 - A. This option provides a correct and concise proof. The argument directly uses the definition of convexity for both sets and shows that the line segment connecting any two points in the intersection remains within the intersection.
 - B. This is incorrect. The intersection of convex sets is always convex.
 - C. This is incorrect. Convexity is defined for sets in R^n for any positive integer n .
 - D. This is incorrect. While the Caratheodory theorem is related to convex sets, it is not necessary to prove this specific statement.
 - E. This is incorrect. Boundedness is not a requirement for the intersection of convex sets to be convex.
 - b. Explain why the convex hull of a set S , denoted $\text{Conv}(S)$, is the smallest convex set containing S .
 - A. This option correctly explains why the definition of $\text{Conv}(S)$ as the intersection of all convex sets containing S implies that it is the smallest such set. Any smaller convex set would necessarily exclude some elements of S , contradicting the definition.
 - B. This is incorrect. The convex hull is defined as the intersection, not the union, of all convex sets containing S .
 - C. While Caratheodory's theorem is related to convex hulls, it doesn't directly explain why $\text{Conv}(S)$ is the smallest convex set containing S .
 - D. This is incorrect. The definition and the property of being the smallest convex set containing S hold regardless of whether S is finite or infinite.
 - E. This is incorrect. The convex hull is defined for any set S , whether convex or not. If S is already convex, then $\text{Conv}(S) = S$.
 - c. Using Caratheodory's Theorem, explain why any point in the convex hull of a set S in R^m can be expressed as a convex combination of at most $m + 1$ points from S .
 - A. While Caratheodory's Theorem states the result, it's crucial to understand the underlying reasoning.
 - B. This option correctly identifies the core idea of the proof of Caratheodory's Theorem, which involves formulating a linear program to find a convex combination representing a given point in the convex hull. The optimal solution of this LP, due to its structure, will involve at most $m + 1$ points.
 - C. The Separation Theorem is a different result in convex analysis and is not directly involved in proving Caratheodory's Theorem.
 - D. This is incorrect. Caratheodory's Theorem applies to any set S , regardless of whether it is finite or infinite.
 - E. This is incorrect. Caratheodory's Theorem applies to any set S , whether it is convex or not. The convex hull is defined for any set.
2. Prove that the intersection of two convex sets is also a convex set.
 - a. This proof correctly demonstrates that the intersection of two convex sets is convex by showing that any convex combination of two points in the intersection remains within the intersection.
 - b. This statement is false. The intersection of convex sets is always convex.

- c. This statement is false. Convexity is defined for any set, including intersections of sets.
 - d. This statement is false. The intersection of two convex sets is always convex, regardless of whether they are disjoint.
 - e. This statement is false. Boundedness is not a requirement for the intersection of convex sets to be convex.
3. Explain why the convex hull of a set S , denoted $\text{Conv}(S)$, is the smallest convex set containing S .
- a. This is incorrect; the convex hull is the smallest, not the largest, convex set containing S .
 - b. This explanation correctly defines the convex hull as the intersection of all convex sets containing S and logically argues why this makes it the smallest such set.
 - c. This is incorrect; the convex hull is always the smallest convex set containing S .
 - d. This is incorrect; the definition and property hold for any set S , whether finite or infinite.
 - e. This is incorrect; the convex hull is defined for any set S , regardless of whether S itself is convex.
4. Using Caratheodory's Theorem, explain why any point in the convex hull of a set S in R^m can be expressed as a convex combination of at most $m + 1$ points from S .
- a. This is incorrect; Caratheodory's Theorem states that at most $m + 1$ points are needed.
 - b. This is incorrect; Caratheodory's Theorem is directly about the number of points in a convex combination representing a point in the convex hull.
 - c. This option correctly states Caratheodory's Theorem and outlines the core idea of its proof using linear programming.
 - d. This is incorrect; the theorem applies to all points in the convex hull, not just interior points.
 - e. This is incorrect; Caratheodory's Theorem provides an upper bound of $m + 1$ points.
5. Consider a set $S = \{(1, 0), (0, 1), (1, 1)\}$ in R^2 . Describe the convex hull of S , $\text{Conv}(S)$, and provide a point within $\text{Conv}(S)$ that is a convex combination of all three points in S .
- a. The description of $\text{Conv}(S)$ is correct, and $(2/3, 2/3)$ is a valid convex combination: $(1/3)(1,0) + (1/3)(0,1) + (1/3)(1,1)$.
 - b. This is incorrect; $\text{Conv}(S)$ includes the entire triangle, not just the line segment.
 - c. This is incorrect; the convex hull is the smallest convex set containing S , which is the triangle.
 - d. This is incorrect; the convex hull is the triangle, not the entire plane.
 - e. This is incorrect; the convex hull is defined for any set.
6. a. Prove that the intersection of two convex sets is also a convex set.
- A. This proof correctly uses the definition of convexity to show that the intersection of two convex sets inherits the convexity property. For any two points in the intersection, their convex combination is also in both sets, and thus in the intersection.
 - B. This is incorrect. The intersection of convex sets is always convex.
 - C. This is incorrect. Convexity is defined for sets in R^n for any positive integer n .

- D. This is incorrect. Disjointness is irrelevant to the convexity of the intersection. The intersection of disjoint sets is simply the empty set, which is itself a convex set.
 - E. This is incorrect. The statement holds true for all convex sets, regardless of their specific type.
- b. Explain why the convex hull of a set S , denoted $\text{Conv}(S)$, is the smallest convex set containing S .
 - A. This explanation correctly uses the definition of the convex hull as the intersection of all convex sets containing S to argue that it is the smallest such set. Any smaller convex set would leave out points of S .
 - B. This is incorrect. The convex hull is the smallest, not the largest, convex set containing S .
 - C. This is incorrect. The definition of the convex hull applies to sets of any cardinality.
 - D. This is incorrect. The convex hull is, by definition, a convex set.
 - E. This is incorrect. The concept of "smallest" is well-defined for sets in terms of set inclusion.
- c. Using Caratheodory's Theorem, explain why any point in the convex hull of a set S in R^m can be expressed as a convex combination of at most $m + 1$ points from S .
 - A. This correctly states Caratheodory's Theorem and indicates that the proof relies on linear programming arguments related to basic feasible solutions.
 - B. This is incorrect. Caratheodory's Theorem applies to any set S , whether convex or not.
 - C. This is incorrect. Caratheodory's Theorem applies to sets of any cardinality.
 - D. This is incorrect. Caratheodory's Theorem states that at most $m + 1$ points are needed.
 - E. This is incorrect. The statement is a direct consequence of Caratheodory's Theorem.